

BIRD ISLAND, South Georgia and the South Sandwich Islands

WMO: 889000

Lat: 54.0086S

Lon: 38.0511W

Elev: 2

StdP: 101.30

Time Zone: -2.00 (W02)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-6.4	-5.2	-10.9	1.5	-5.7	-9.6	1.7	-4.6	15.0	1.1	13.8	-0.9	6.7	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.5	7.1	6.0	6.4	5.5	5.9	5.0	6.4	6.9	5.7	6.2	5.2	5.7	5.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.9	5.7	6.6	5.3	5.5	6.0	4.7	5.3	5.5	21.6	7.0	20.3	6.4	19.3	5.8	8.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.4	10.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-8.6	7.8	1.3	0.7	-9.6	8.2	-10.3	8.6	-11.1	9.0	-12.1	9.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	3.6	3.7	3.4	2.2	0.8	0.1	-0.8	-0.8	-0.1	0.8	1.5	2.7		
	DBStd	2.30	1.26	1.36	1.36	1.71	2.12	1.77	2.19	2.07	1.71	1.26	1.40	1.19		
	HDD10.0	3127	200	176	203	233	284	297	333	334	302	285	254	225		
	HDD18.3	6169	458	409	462	483	542	547	592	593	552	543	504	483		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	5.1	4.7	5.0	5.2	5.4	5.5	5.2	5.3	5.4	5.4	5.0	4.6	4.7		
	PrecAvg	1404	107	119	150	140	136	117	116	118	119	108	93	82		
(15) Precipitation	PrecMax	1648	169	186	222	241	204	164	186	186	254	172	134	128		
	PrecMin	1119	63	67	118	95	90	76	71	75	36	73	44	45		
	PrecStd	163	29	27	26	29	24	22	26	25	35	23	18	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.6	7.9	6.8	5.7	4.2	4.0	3.7	4.5	4.6	5.8	6.8		
		MCWB	6.8	6.6	6.8	5.4	4.7	3.4	3.2	2.9	3.1	3.3	4.2	5.4		
	2%	DB	6.7	7.1	6.8	5.7	5.0	3.5	3.3	2.9	3.2	3.7	4.9	5.7		
		MCWB	5.7	5.9	5.9	4.9	4.3	2.8	2.6	2.1	2.3	2.8	3.9	4.7		
	5%	DB	6.0	6.3	6.1	5.2	4.4	3.0	2.7	2.3	2.6	3.2	4.2	5.1		
		MCWB	5.0	5.4	5.3	4.5	3.7	2.4	2.0	1.6	1.8	2.4	3.4	4.2		
	10%	DB	5.4	5.6	5.5	4.6	3.8	2.4	2.2	1.9	2.1	2.7	3.6	4.6		
		MCWB	4.5	4.8	4.8	3.9	3.1	1.7	1.5	1.2	1.4	2.0	2.8	3.8		
	0.4%	WB	7.0	6.9	7.3	5.8	5.2	3.7	3.6	3.3	3.6	3.7	5.0	5.8		
		MCDB	7.6	7.6	7.8	6.0	5.4	3.9	3.9	3.6	4.2	4.1	5.5	6.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	6.0	6.3	6.2	5.2	4.6	3.1	3.0	2.5	2.6	3.1	4.1	5.0		
		MCDB	6.5	6.8	6.7	5.6	4.9	3.3	3.3	2.9	3.0	3.6	4.7	5.5		
	5%	WB	5.2	5.7	5.6	4.7	4.1	2.6	2.3	1.9	2.1	2.6	3.6	4.4		
		MCDB	5.7	6.2	6.1	5.1	4.4	2.9	2.7	2.3	2.5	3.1	4.1	5.0		
	10%	WB	4.7	5.0	5.0	4.1	3.3	2.0	1.7	1.4	1.6	2.2	3.1	4.0		
		MCDB	5.2	5.6	5.5	4.6	3.8	2.4	2.1	1.8	2.1	2.7	3.6	4.5		
	5% DB	MDBR	2.4	2.5	2.5	2.6	2.6	2.5	2.7	2.7	2.5	2.3	2.3	2.4		
		MCDBR	3.0	3.5	3.2	3.4	2.8	2.9	3.0	2.6	3.1	2.9	3.0	3.1		
		MCWBR	2.7	2.9	3.1	3.4	2.9	2.9	3.1	2.6	2.9	2.7	2.6	2.7		
		MCWBR	2.8	3.3	3.1	3.3	2.8	2.9	2.9	2.5	3.1	2.9	3.0	3.0		
(35) Clear-Sky Solar Irradiance	taub		0.342	0.339	0.330	0.315	0.293	0.280	0.291	0.310	0.323	0.330	0.327	0.333		
		taud	2.479	2.491	2.508	2.467	2.359	2.273	2.307	2.376	2.417	2.441	2.489	2.489		
	Ebn at Noon		957	921	864	763	639	560	621	749	853	920	968	977		
		Edh at Noon	107	97	83	67	52	45	53	71	89	101	105	108		
(44) All-Sky Solar Radiation	RadAvg		4.51	3.67	2.34	1.32	0.65	0.41	0.51	1.02	1.99	3.20	4.45	5.10		
	RadStd		0.31	0.26	0.16	0.11	0.05	0.03	0.04	0.07	0.16	0.18	0.31	0.33		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page